

1. Introduction

Probability theory is a fundamental area of mathematics with wide applications in engineering, statistics, data science, and many applied sciences. Many important results in probability describe the behavior of random variables when the number of observations becomes very large. However, these theoretical results are often abstract and difficult to fully understand using only mathematical formulas. For this reason, computer simulations and numerical experiments play a critical role in developing an intuitive understanding of probabilistic concepts.

Team Work 5 (TW5) is the final stage of a sequence of simulation-based projects conducted in the IE 221 – Probability course. This sequence includes TW3, TW4, and TW5, each of which builds upon the previous one. In TW3, the Strong Law of Large Numbers (SLLN) and the Central Limit Theorem (CLT) were introduced and experimentally verified using basic Monte Carlo simulations. These simulations helped demonstrate how sample means and distributions behave as the sample size increases.

TW4 extended this foundation by providing a more detailed theoretical background. In that phase, different modes of convergence were discussed, such as almost sure convergence and convergence in distribution. Additionally, the Monte Carlo method was studied in depth, and simulations were used to illustrate how probabilistic theorems support numerical estimation techniques. TW4 emphasized the importance of understanding both the theory and the experimental methodology behind probabilistic convergence.

Building directly on the theoretical framework and simulation methodology developed in TW3 and TW4, the main objective of TW5 is to explore the limits of the Strong Law of Large Numbers and the Central Limit Theorem. While these theorems are often presented as general results, they rely on specific assumptions, such as the existence of finite moments. In this phase, these assumptions are tested by applying SLLN and CLT to distributions with very different properties, including skewed, heavy-tailed, and non-standard distributions.

In particular, this study focuses on how tail behavior and the existence of moments affect convergence. Through Monte Carlo simulations, this report investigates cases where convergence is fast and smooth, cases where convergence is slow and noisy, and cases where convergence fails completely. By comparing simulation results across different distributions, TW5 aims to provide a clear and practical understanding of when classical probabilistic theorems can be safely applied and when caution is required.

2. Introduction to Distributions

In this study, five different probability distributions are analyzed: Uniform(0,1), Exponential($\lambda = 1$), Pareto($\alpha = 3$), Pareto($\alpha = 1.5$), and the Cauchy distribution. These distributions were selected because they exhibit significantly different characteristics in terms of symmetry, skewness, tail behavior, and existence of moments.

Understanding the expected value and variance of each distribution is essential for interpreting the simulation results. In particular, the validity of the Strong Law of Large Numbers requires the existence of a finite expected value, while the Central Limit Theorem additionally requires a finite variance.

Table 1: Moments of the Distributions

DISTRIBUTION	$E[X]$	$VAR(X)$
UNIFORM(0,1)	0.5	$1/12 \approx 0.0833$
EXPONENTIAL($\Lambda = 1$)	1	1
PARETO($\Lambda = 3, X_m = 1$)	1.5	0.75
PARETO($\Lambda = 1.5, X_m = 1$)	∞ (undefined)	∞ (undefined)
CAUCHY(0,1)	undefined	undefined

The Uniform and Exponential distributions are light-tailed and have finite mean and variance. The Pareto distribution with $\alpha = 3$ has heavier tails but still satisfies the necessary moment conditions. In contrast, Pareto($\alpha = 1.5$) has infinite variance, and the Cauchy distribution has neither a defined mean nor variance. These differences are expected to strongly influence the validity of SLLN and CLT.

3. Strong Law of Large Numbers (SLLN) Results

To investigate the Strong Law of Large Numbers, at least 10,000 independent observations were generated for each distribution. The cumulative sample mean was computed and plotted as a function of the sample size in order to observe convergence behavior.

3.1 Uniform(0,1)

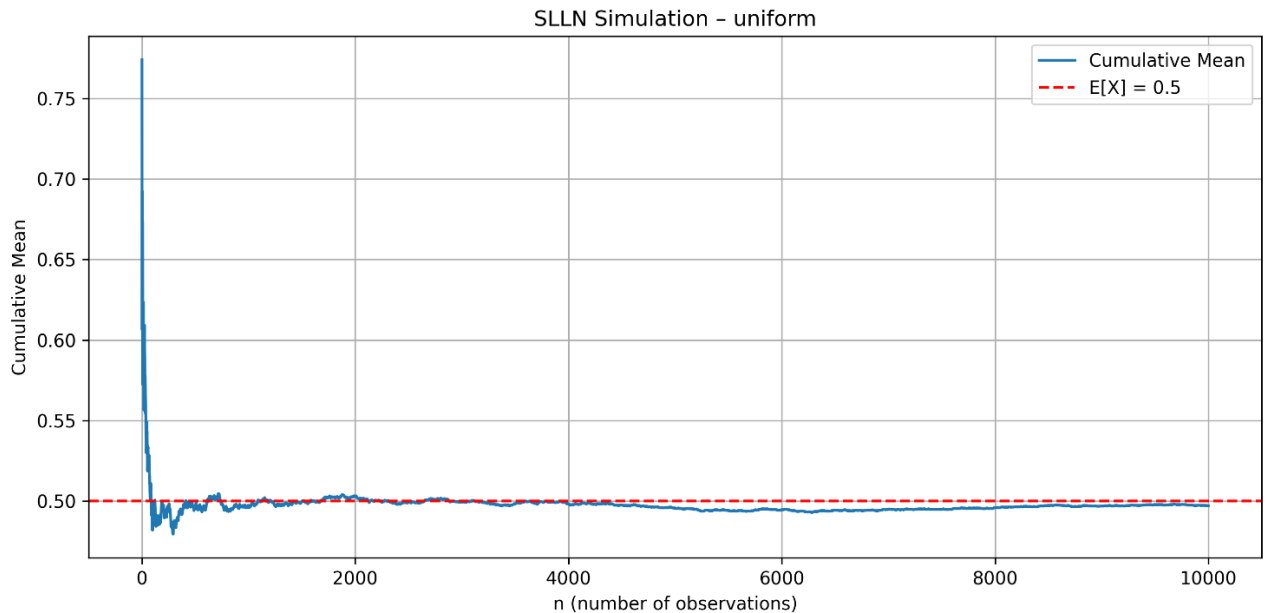


Figure 1: Uniform(0,1)

The cumulative sample mean converges smoothly to the theoretical mean value of 0.5. As the sample size increases, fluctuations become smaller and the mean stabilizes. This behavior provides clear experimental evidence that the Strong Law of Large Numbers holds for the Uniform(0,1) distribution.

3.2 Exponential($\lambda = 1$)

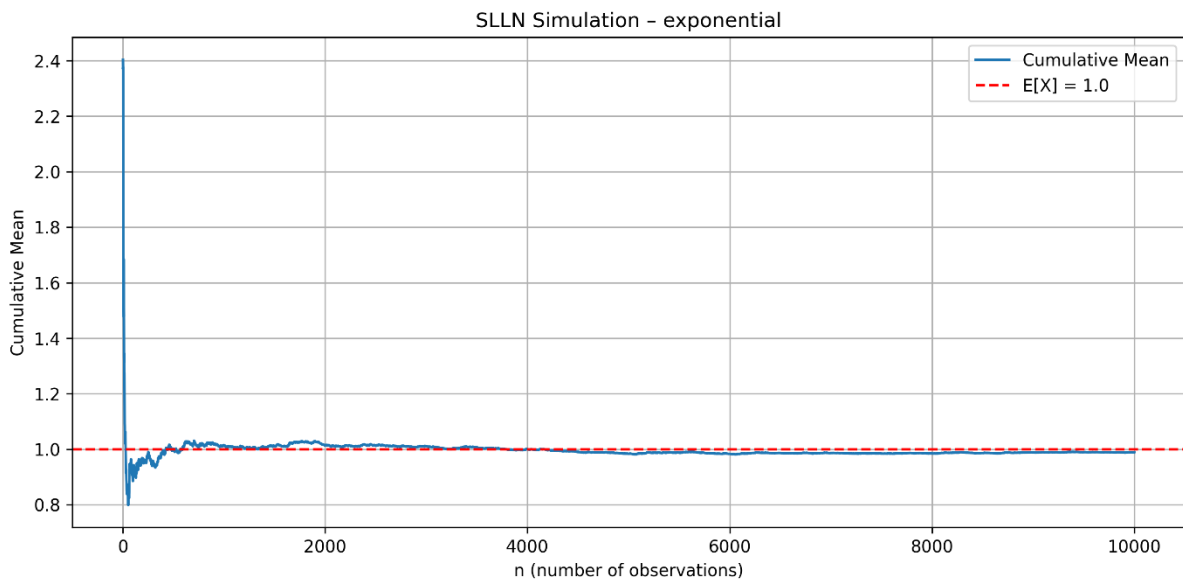


Figure 2: Cumulative mean of Exponential(1) random variables.

For the Exponential(1) distribution, the sample mean converges to the theoretical mean value of 1. Compared to the uniform case, convergence is slower due to the strong right skewness of the distribution. However, the mean still stabilizes, confirming that SLLN holds.

3.3 Pareto Distribution ($\alpha = 3$)

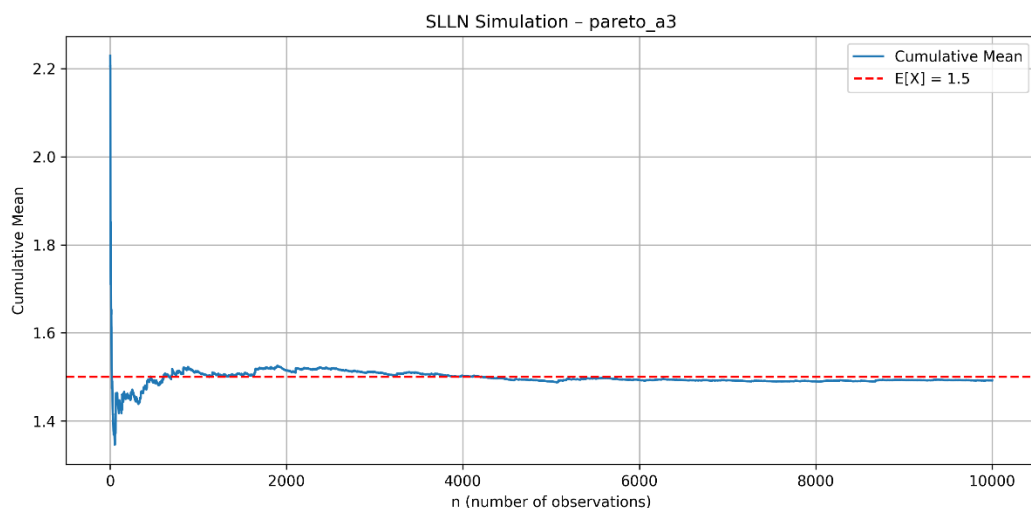


Figure 3: Cumulative mean of Pareto($\alpha = 3$) random variables.

The cumulative mean converges to the theoretical value of 1.5, but the convergence is noisy and shows large fluctuations. This behavior is caused by the heavy-tailed nature of the Pareto

distribution. Despite these fluctuations, the existence of a finite expected value ensures that SLLN remains valid.

3.4 Pareto Distribution ($\alpha = 1.5$)

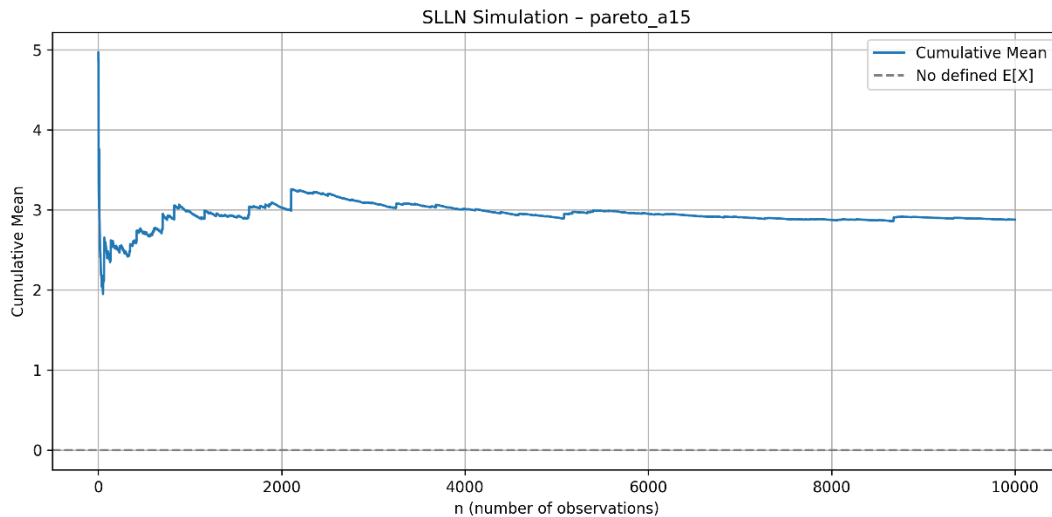


Figure 4: Cumulative mean of Pareto($\alpha = 1.5$) random variables.

The cumulative mean does not stabilize and exhibits large jumps even for large sample sizes. Since the expected value of this distribution is infinite, the conditions of the Strong Law of Large Numbers are not satisfied. As a result, SLLN does not hold.

3.5 Cauchy Distribution

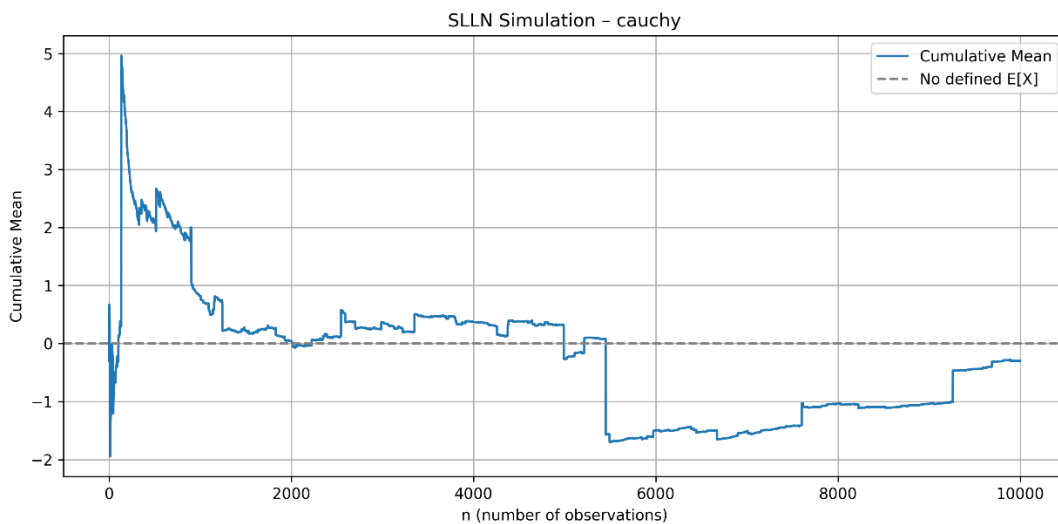


Figure 5: Cumulative mean of Cauchy random variables.

The sample mean behaves chaotically and never converges to a stable value. The Cauchy distribution has no defined mean, and therefore the Strong Law of Large Numbers does not apply. This example demonstrates a complete failure of SLLN.

4. Central Limit Theorem (CLT) Results

The Central Limit Theorem was investigated using Monte Carlo simulations. For each distribution, standardized sums were generated for sample sizes $n \in \{2, 5, 10, 30, 50, 100\}$ with $m = 1000$ replications. Histograms and Normal Q–Q plots were used to assess convergence to the standard normal distribution.

4.1 Uniform(0,1)

The histograms rapidly approach a bell-shaped normal distribution as the sample size increases. The Q–Q plots align closely with the reference line for $n \geq 30$, indicating fast convergence. These results clearly confirm the validity of the Central Limit Theorem.

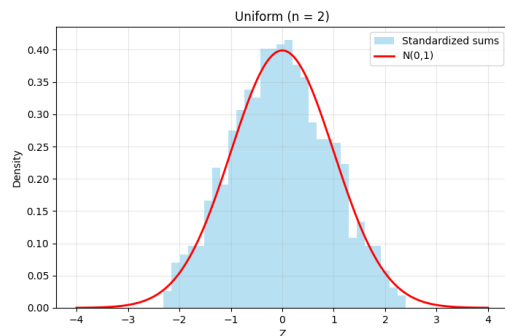


Figure 6: Uniform($n = 2$)

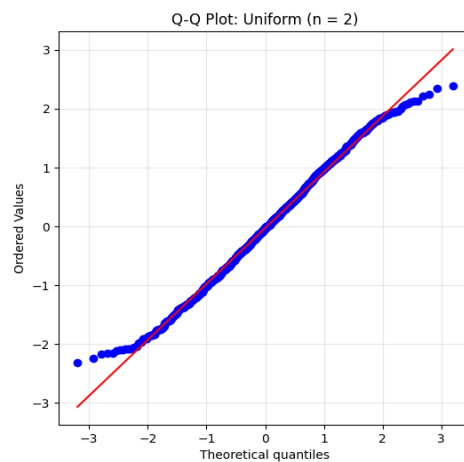


Figure 7: Q-Q Plot Uniform ($n=2$)

Explanation for $n = 2$

For $n = 2$, the Central Limit Theorem is not yet observable. The histogram of the standardized sums shows a clearly non-normal shape, and the distribution is far from bell-shaped. The Q–Q plot displays strong deviations from the reference line, especially at the tails. These results indicate that the sample size is too small for convergence to the normal distribution.

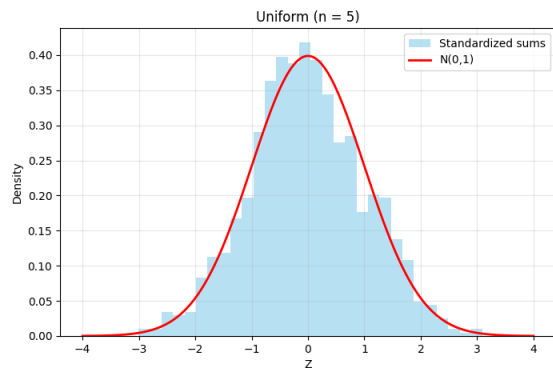


Figure 8:Uniform(n=5)

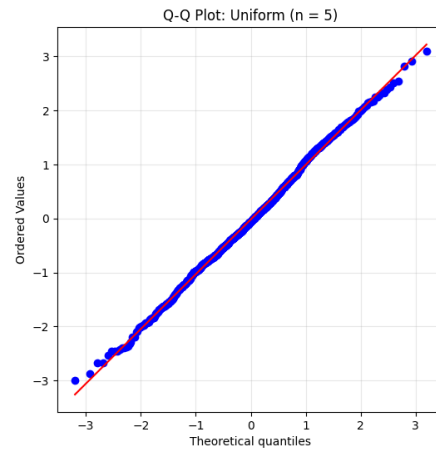


Figure 9:Q-Q Plot Uniform (n=2)

Explanation for $n = 5$

When the sample size increases to $n = 5$, the effect of averaging begins to appear. The histogram becomes slightly more symmetric compared to $n = 2$, but noticeable deviations from the normal shape remain. The Q–Q plot still shows curvature and deviations from the reference line, indicating that convergence has started but is still weak.

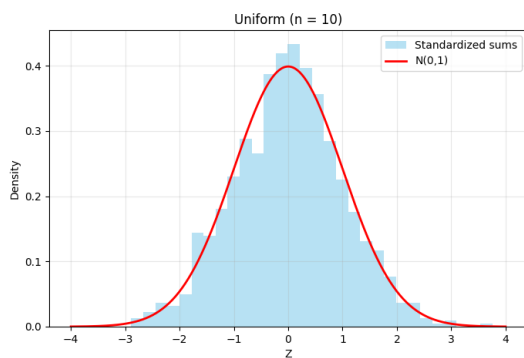


Figure 10:Uniform(n=10)

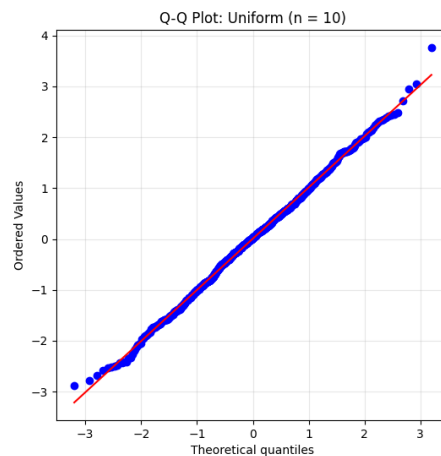


Figure 11:Q-Q Plot Uniform(n=10)

Explanation for $n = 10$

At $n = 10$, the convergence behavior becomes more visible. The histogram shows a clearer bell-shaped structure, and extreme irregularities are reduced. However, the Q–Q plot still exhibits small deviations from the reference line, particularly at the tails. At this stage, the normal approximation is improving but not yet fully accurate.

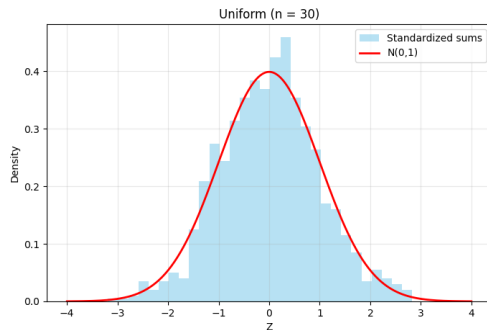


Figure 12:Uniform(n=30)

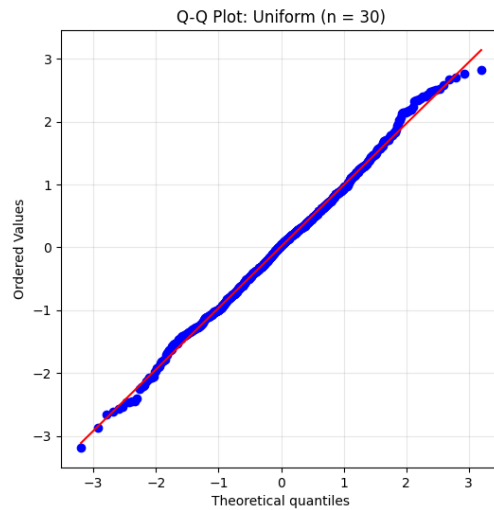


Figure 13:Q-Q Plot Uniform(n=30)

Explanation for $n = 30$

For $n = 30$, the Central Limit Theorem becomes clearly evident.

The histogram closely resembles the bell-shaped normal distribution, and the Q–Q plot aligns well with the reference line for most quantiles. Only minor deviations are observed at the extreme tails. This sample size provides a good approximation to the normal distribution.

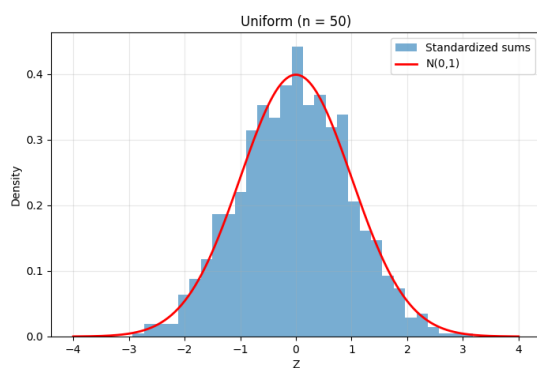


Figure 14:Uniform(n=50)

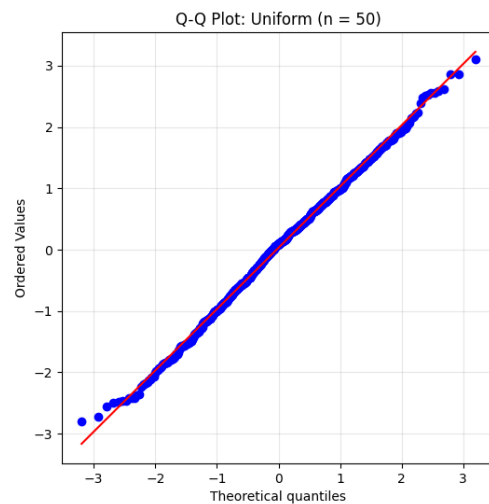


Figure 15:Q-Q Plot Uniform(n=50)

Explanation for $n = 50$

When the sample size reaches $n = 50$, convergence to the normal distribution becomes stronger and more stable.

The histogram is smooth and symmetric, and the Q–Q plot follows the reference line closely. Deviations are minimal and mostly limited to the tails. These results indicate that the normal approximation is very accurate for this sample size.

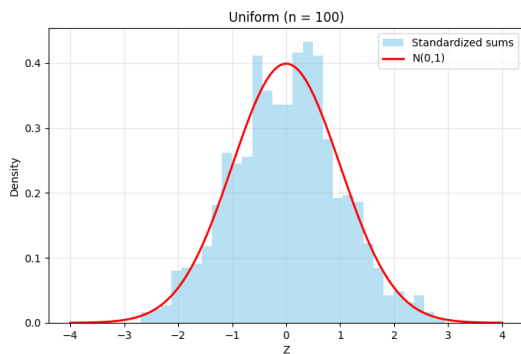


Figure 16:Uniform(n=100)

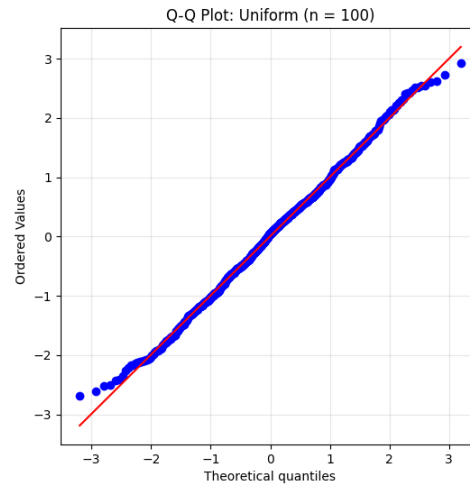


Figure 17 :Q-Q Plot Uniform(n=100)

Explanation for $n = 100$

For $n = 100$, the Central Limit Theorem is strongly supported.

The histogram is almost indistinguishable from the standard normal distribution, and the Q–Q plot points lie nearly perfectly on the reference line. Any remaining deviations are negligible. This result demonstrates that a sufficiently large sample size leads to strong convergence in distribution for the Uniform(0,1) distribution.

4.2 Exponential($\lambda = 1$)

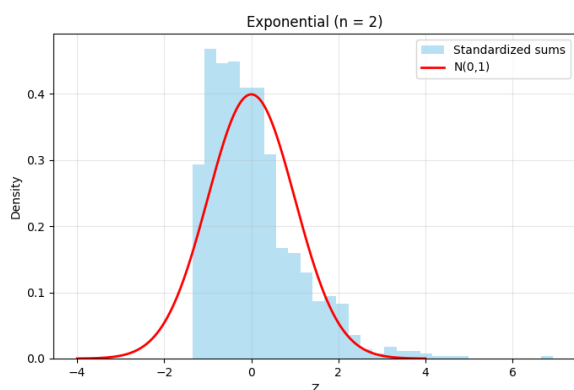


Figure 18:Exponential(n = 2)

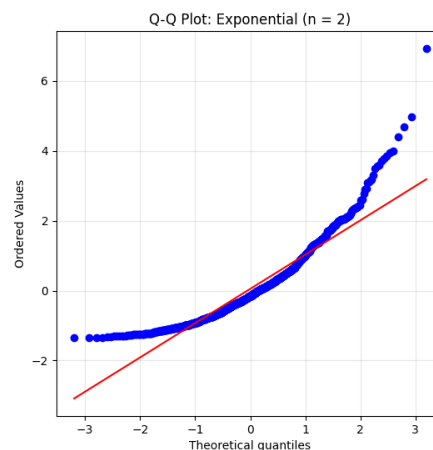


Figure 19: Q-Q Plot Exponential (n=2)

Explanation for $n = 2$

For $n = 2$, the Central Limit Theorem is not clearly observed. The histogram of the standardized sums is highly skewed and does not resemble a normal distribution. The Q-Q plot shows strong curvature and large deviations from the reference line, especially in the upper tail. This behavior is expected due to the strong right-skewness of the exponential distribution and the very small sample size.

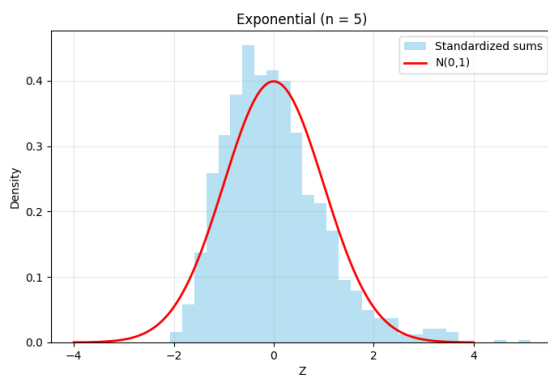


Figure 20: Exponential($n=5$)

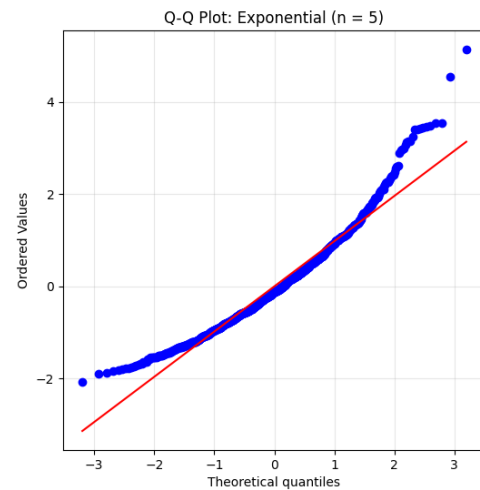


Figure 21: Q-Q Plot Exponential ($n=5$)

Explanation for $n = 5$

When the sample size increases to $n = 5$, partial averaging effects appear. The histogram becomes slightly more symmetric compared to $n = 2$, but clear skewness is still present. The Q-Q plot shows noticeable deviations from the reference line, indicating that convergence to normality has started but remains weak at this stage.

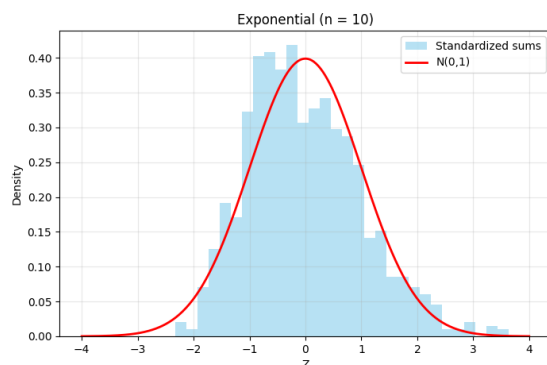


Figure 22: Uniform($n=10$)

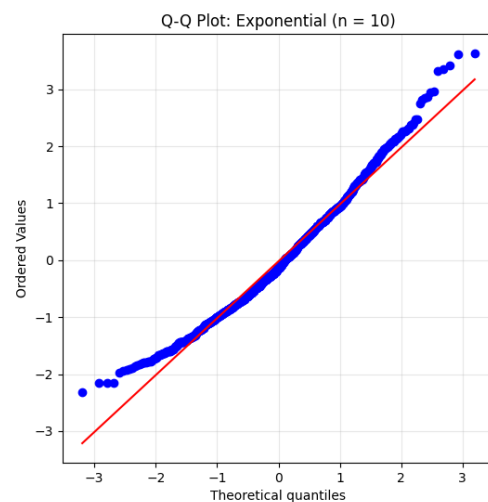


Figure 23: Q-Q Plot Uniform($n=10$)

Explanation for $n = 10$

At $n = 10$, the distribution of the standardized sums improves.

The histogram shows a more bell-shaped form, although some right-skewness is still visible. The Q-Q plot follows the reference line more closely than for smaller sample sizes, but tail deviations remain. The normal approximation is improving but is not yet fully accurate.

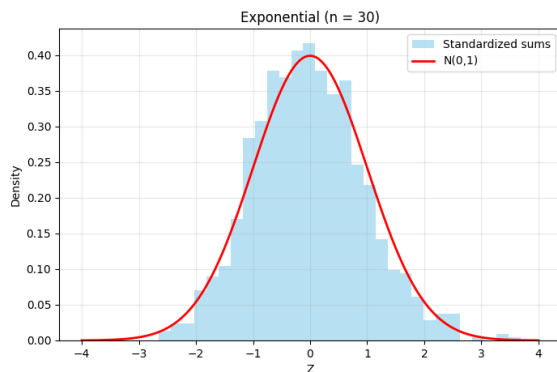


Figure 24: Exponential($n=30$)

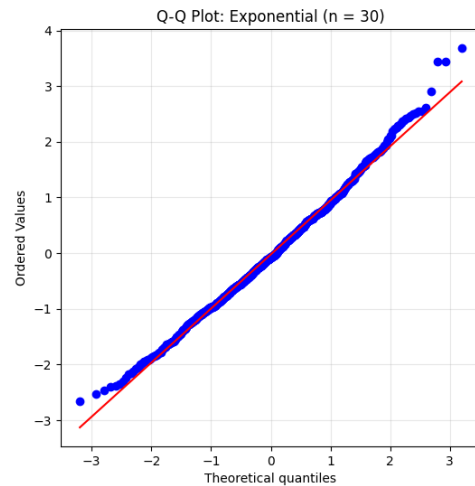


Figure 25: Q-Q Plot Exponential($n=30$)

Explanation for $n = 30$

For $n = 30$, convergence to the normal distribution becomes clear.

The histogram closely resembles a bell-shaped curve, and the Q-Q plot aligns well with the reference line for most quantiles. Minor deviations may still appear at the upper tail due to skewness, but overall the normal approximation is reasonably accurate.

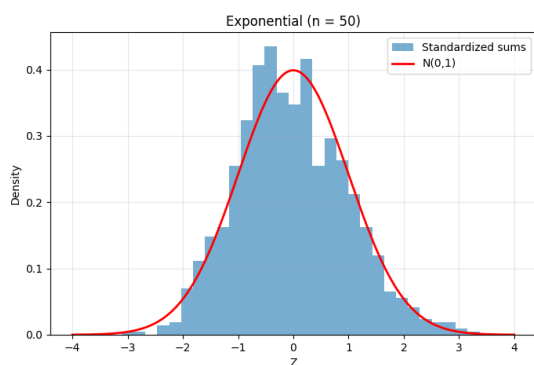


Figure 26: Exponential($n=50$)

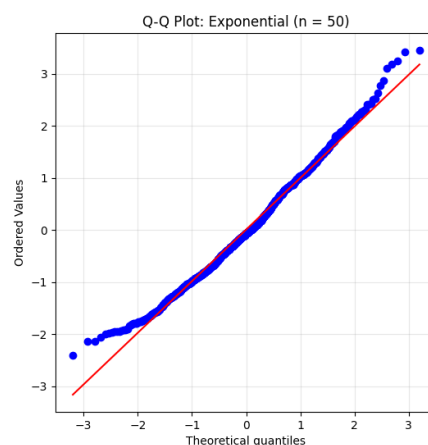


Figure 27: Q-Q Plot Exponential($n=50$)

Explanation for $n = 50$

When the sample size reaches $n = 50$, convergence becomes stronger and more stable. The histogram is smoother and more symmetric, and the Q-Q plot follows the reference line closely with only small deviations at the tails. These results indicate that the effect of skewness is significantly reduced at this sample size.

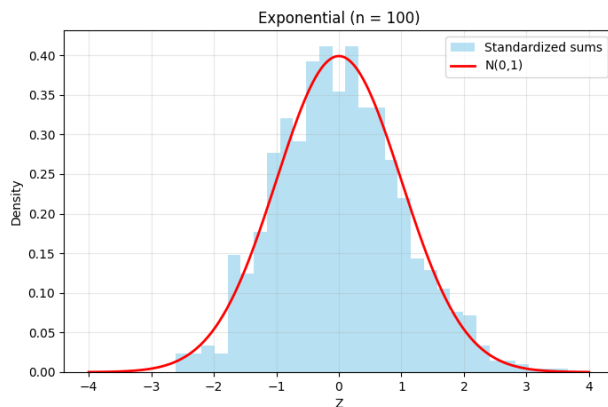


Figure 28: Exponential($n=100$)

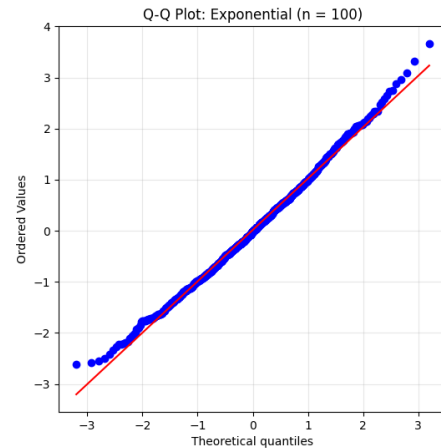


Figure 29 : Q-Q Plot Exponential($n=100$)

Explanation for $n = 100$

For $n = 100$, the Central Limit Theorem is strongly supported. The histogram is very close to the standard normal distribution, and the Q-Q plot points lie almost entirely on the reference line. Remaining deviations are minimal. This result confirms that, despite strong skewness in the original distribution, a sufficiently large sample size leads to convergence in distribution.

4.3 Pareto Distribution ($\alpha = 3$)

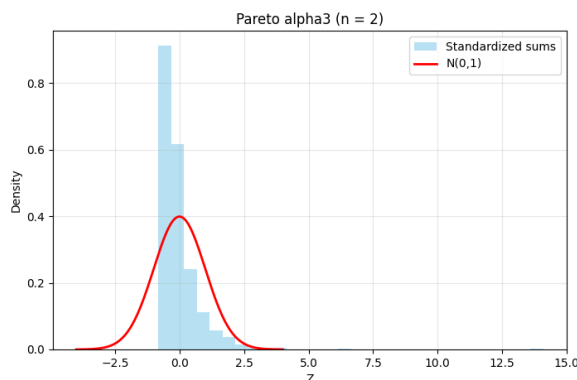


Figure 30: Pareto($n = 2$)

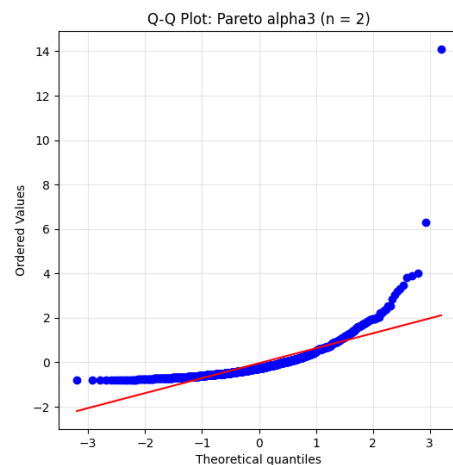


Figure 31: Q-Q Plot Pareto ($n=2$)

Explanation for $n = 2$

For $n = 2$, the Central Limit Theorem is not observable.

The histogram shows a highly irregular and heavy-tailed shape, and it is far from a bell-shaped distribution. The Q–Q plot displays strong deviations from the reference line, especially at the upper tail. This behavior is caused by the heavy-tailed nature of the Pareto distribution and the very small sample size.

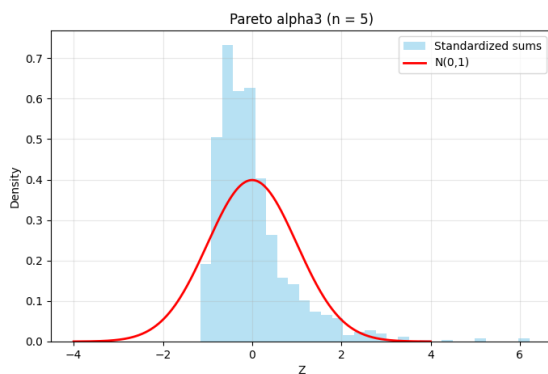


Figure 32: Pareto($n=5$)

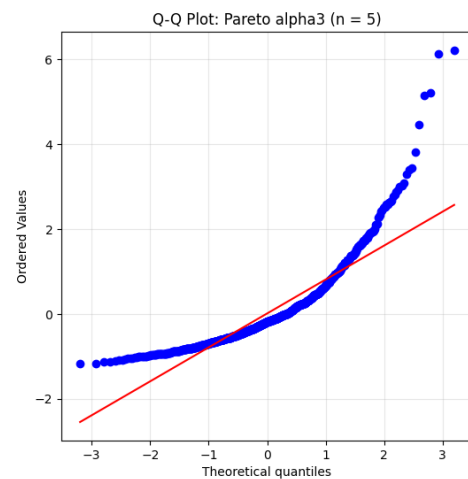


Figure 33: Q-Q Plot Pareto ($n=5$)

Explanation for $n = 5$

When the sample size increases to $n = 5$, some averaging effects appear, but convergence remains weak. The histogram becomes slightly more symmetric compared to $n = 2$, but heavy tails are still clearly visible. The Q–Q plot shows noticeable curvature and large tail deviations, indicating slow convergence toward normality.

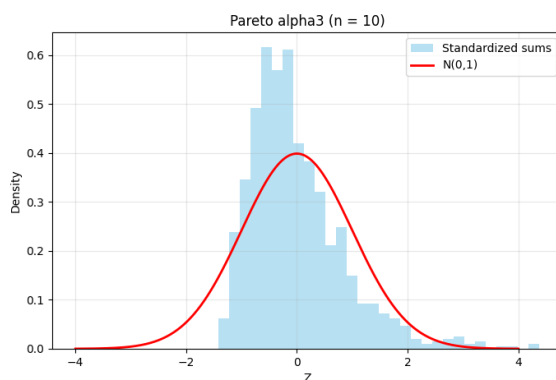


Figure 34: Pareto($n=10$)

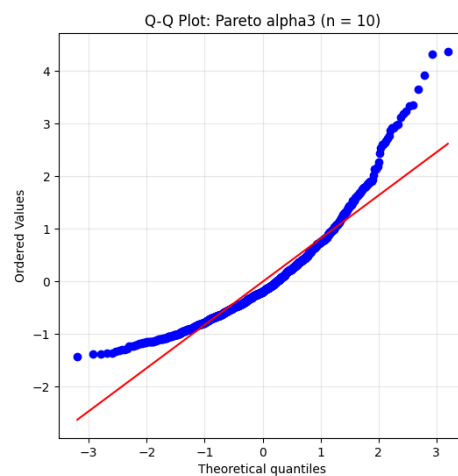


Figure 35: Q-Q Plot Pareto($n=10$)

Explanation for $n = 10$

At $n = 10$, partial convergence to the normal distribution can be observed.

The histogram begins to show a rough bell-shaped structure, but extreme values still affect the shape. The Q–Q plot follows the reference line more closely than for smaller sample sizes, yet clear deviations remain at the tails. The normal approximation is improving but remains limited due to heavy tails.

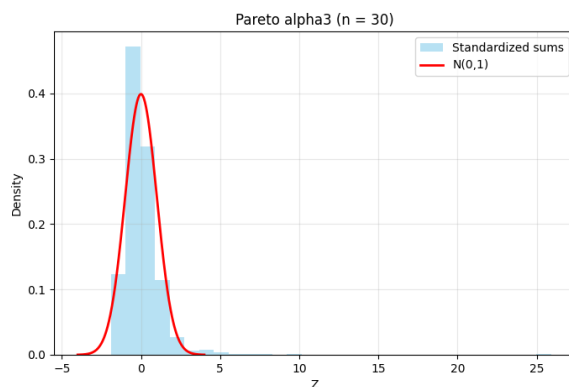


Figure 36: Pareto($n=30$)

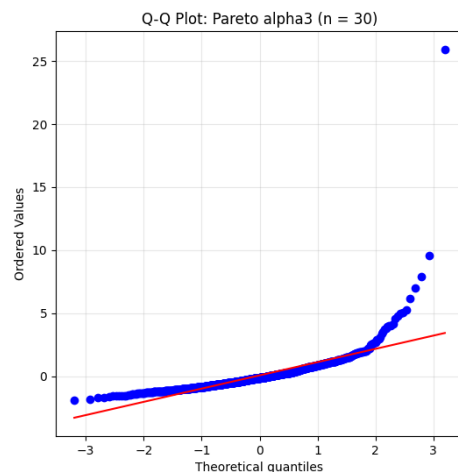


Figure 37: Q-Q Plot Pareto($n=30$)

Explanation for $n = 30$

For $n = 30$, convergence becomes more noticeable but is slower than for light-tailed distributions. The histogram shows a clearer bell-shaped form, but heavier tails are still present. The Q–Q plot aligns reasonably well with the reference line in the center, while deviations persist at the extremes. This indicates moderate convergence to normality.

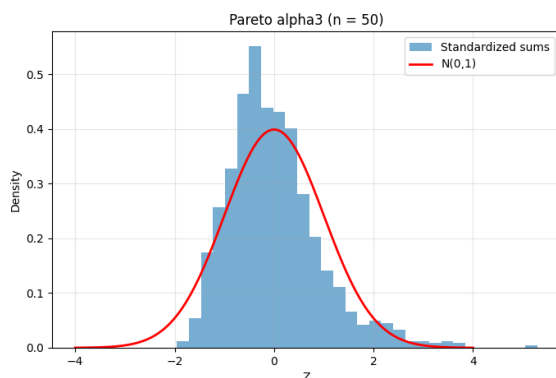


Figure 38: Pareto ($n=50$)

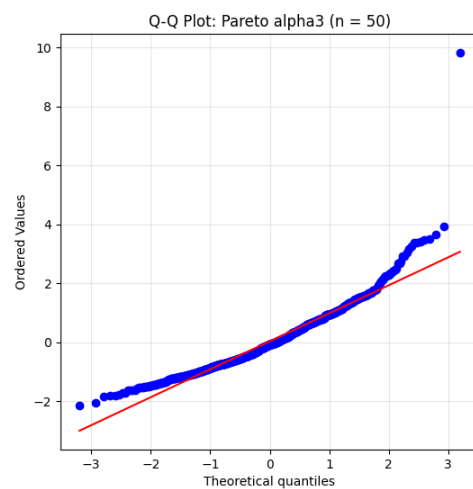


Figure 39: Q-Q Plot Pareto($n=50$)

Explanation for $n = 50$

When the sample size reaches $n = 50$, convergence continues to improve. The histogram becomes smoother and more symmetric, although tail heaviness is still visible. The Q–Q plot follows the reference line closely in the central region, with remaining deviations at the tails. These results show that larger sample sizes are required for heavy-tailed distributions to achieve good normal approximation.

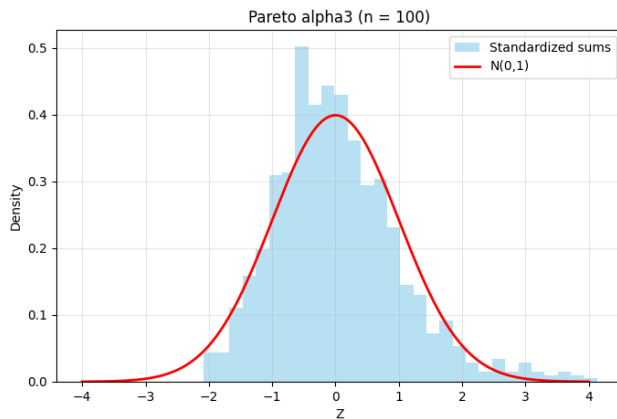


Figure 40: Pareto($n=100$)

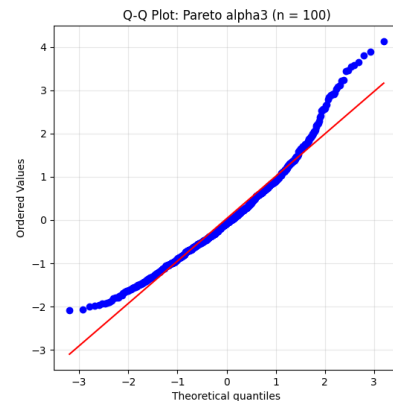


Figure 41 : Q-Q Plot Pareton= 100)

Explanation for $n = 100$

For $n = 100$, the Central Limit Theorem is largely supported, but convergence is still slower compared to light-tailed distributions. The histogram is close to a bell-shaped normal distribution, yet slightly heavier tails remain. The Q–Q plot aligns well with the reference line for most quantiles, with minor deviations at the extremes. This confirms that the CLT holds for Pareto($\alpha = 3$), but large sample sizes are needed due to heavy-tailed behavior.

4.3 Pareto Distribution ($\alpha = 1/5$)

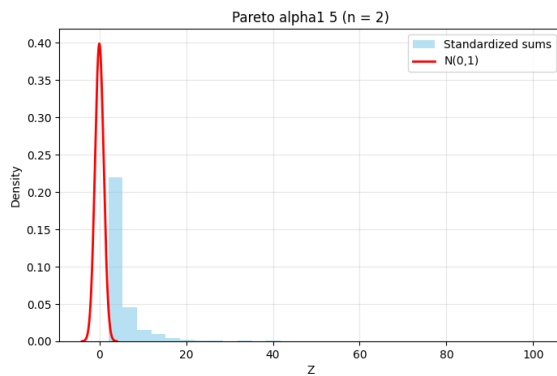


Figure 42: Pareto($n=2$)

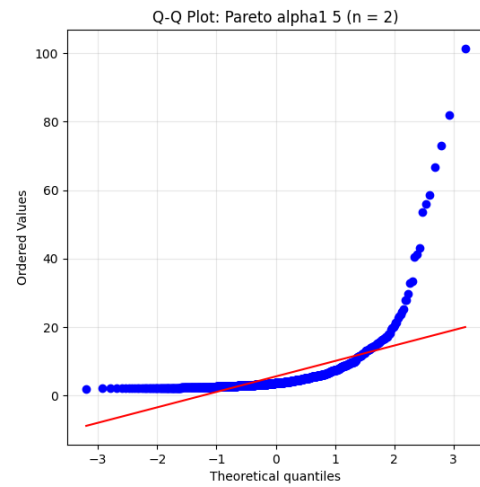


Figure 43: Q-Q Plot Pareto ($n=2$)

Explanation for $n = 2$

For $n = 2$, the distribution of the standardized sums is highly irregular and heavy-tailed. The histogram is far from a bell-shaped form, and extreme values strongly affect the shape. The Q-Q plot shows large deviations from the reference line, especially at the upper tail. No convergence to the normal distribution is observed

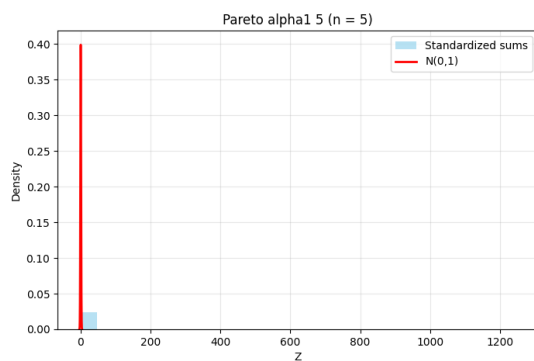


Figure 44: Pareto($n=5$)

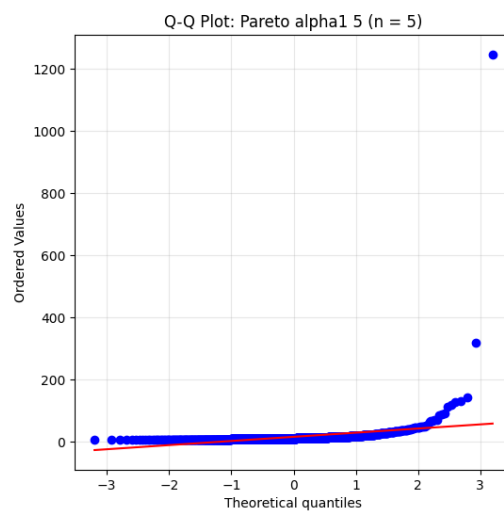


Figure 45: Q-Q Plot Pareto ($n=5$)

Explanation for $n = 5$

When the sample size increases to $n = 5$, some averaging effects appear, but the distribution remains heavy-tailed.

The histogram still shows strong asymmetry and extreme observations. The Q-Q plot displays severe curvature and large deviations from linearity. These results indicate that increasing the sample size does not lead to normal convergence.

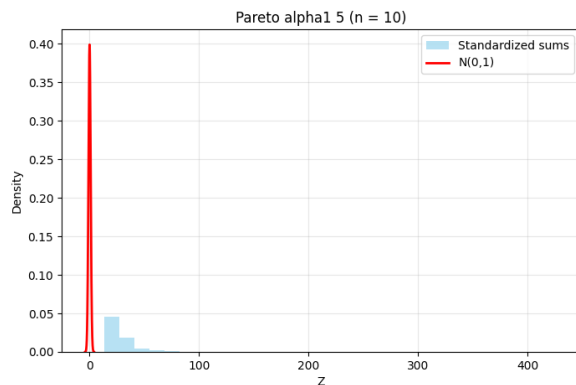


Figure 46: Pareto($n=10$)

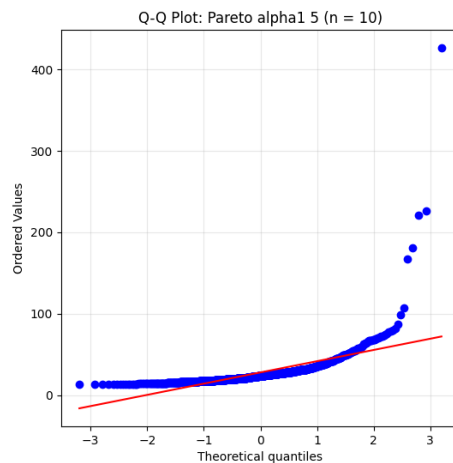


Figure 47: Q-Q Plot Pareto($n=10$)

Explanation for $n = 10$

At $n = 10$, the standardized sums continue to exhibit non-normal behavior.

The histogram remains irregular, and heavy tails dominate the distribution. The Q-Q plot shows persistent and strong deviations from the reference line across most quantiles. The Central Limit Theorem is not observed at this sample size.

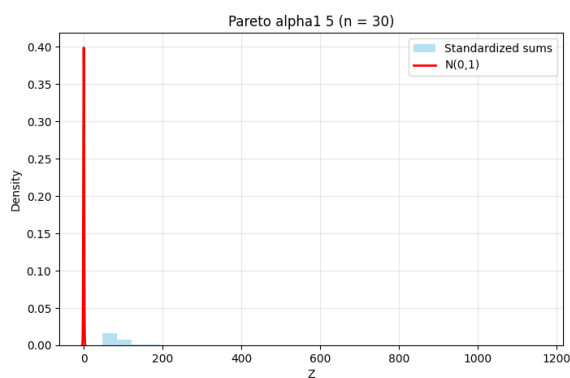


Figure 48: Pareto($n=30$)

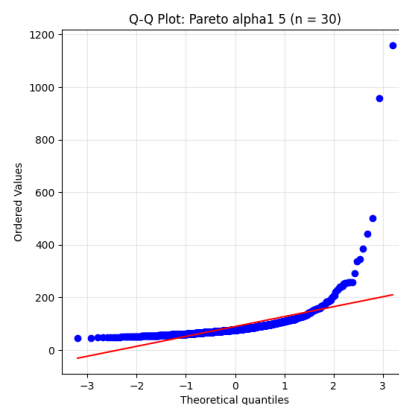


Figure 49: Q-Q Plot Pareto($n=30$)

Explanation for $n = 30$

For $n = 30$, the lack of convergence becomes more evident. Despite the larger sample size, the histogram does not approach a bell-shaped form and remains heavy-tailed. The Q-Q plot shows extreme deviations, particularly at the tails. This confirms that increasing n does not improve normality for this distribution.

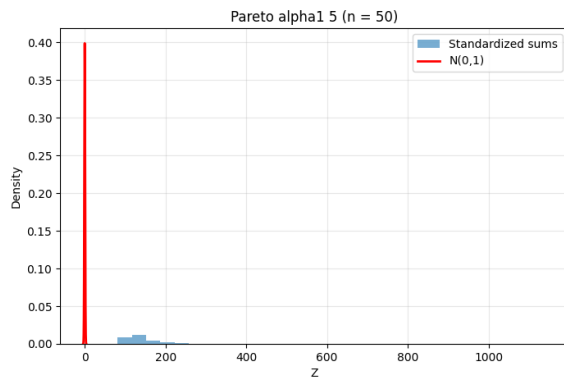


Figure 50: Pareto ($n=50$)

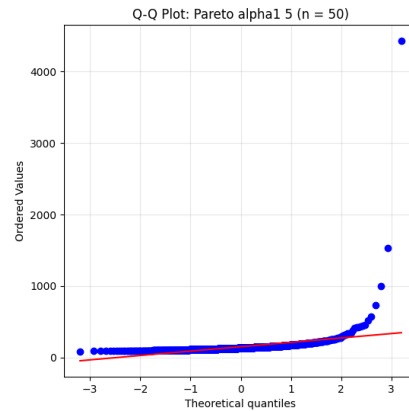


Figure 51: Q-Q Plot Pareto($n=50$)

Explanation for $n = 50$

When the sample size reaches $n = 50$, the behavior remains largely unchanged. The histogram continues to show heavy tails and irregular shape, and extreme values still dominate the distribution. The Q-Q plot deviates strongly from the reference line. These results highlight the failure of the Central Limit Theorem due to infinite variance.

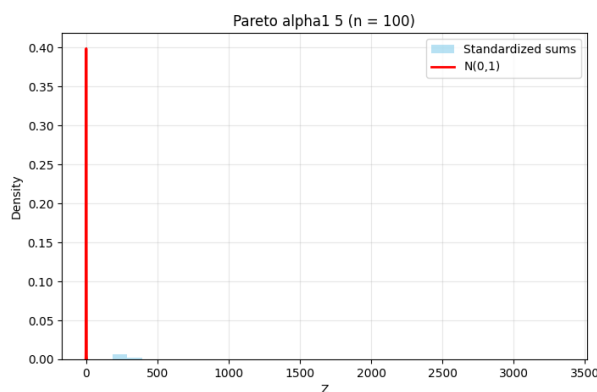


Figure 52: Pareto($n=100$)

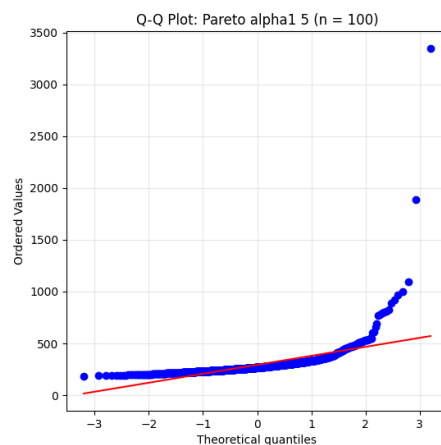


Figure 53 : Q-Q Plot Pareton= 100)

Explanation for $n = 100$

For $n = 100$, there is still no convergence toward the normal distribution. The histogram remains heavy-tailed and does not resemble a normal shape. The Q-Q plot

shows severe deviations from linearity across all quantiles. This confirms that the Central Limit Theorem fails completely for the Pareto($\alpha = 1.5$) distribution, even for large sample sizes.

4.4 Cauchy Distribution

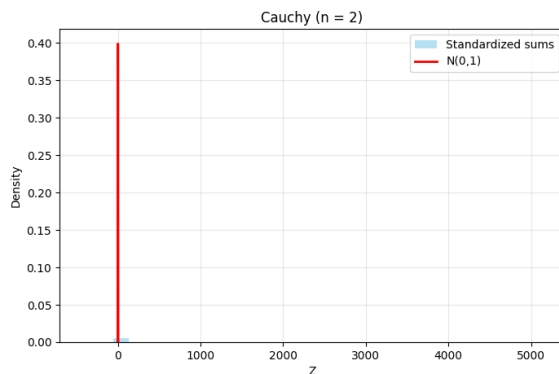


Figure 54:Cauchy($n = 2$)

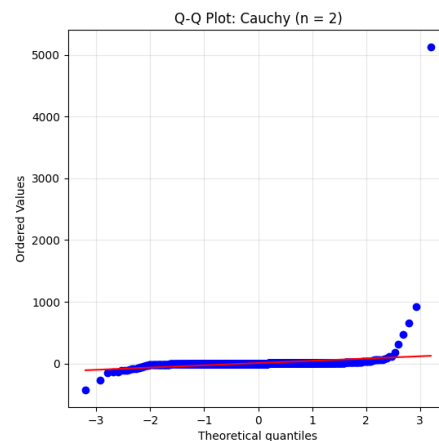


Figure 55: Q-Q Plot Cauchy ($n=2$)

Explanation for $n = 2$

For $n = 2$, the standardized sums exhibit a highly irregular and unpredictable distribution. The histogram is far from a bell-shaped curve, showing a highly peaked and heavy-tailed shape. The Q–Q plot demonstrates extreme deviations from the reference line across almost all quantiles. This behavior is due to the undefined mean and variance of the Cauchy distribution, and no convergence to normality is observed.

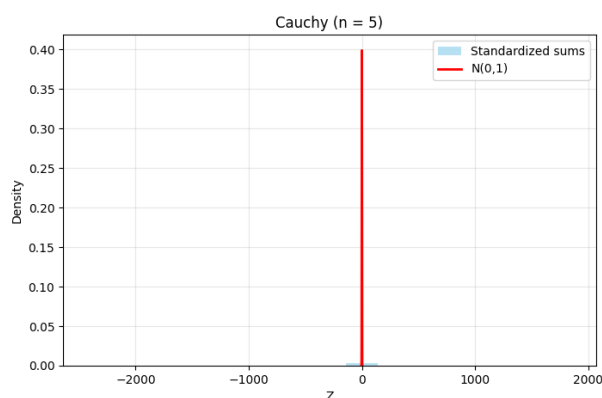


Figure 56 Cauchy ($n=5$)

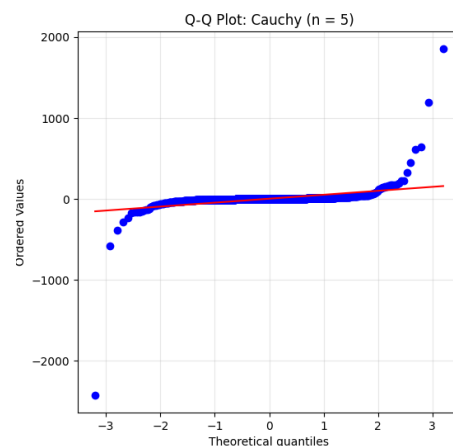


Figure 57:Q-Q Plot Cauchy ($n=5$)

Explanation for $n = 5$

When the sample size increases to $n = 5$, the distribution of the standardized sums remains highly irregular.

The histogram still shows a heavy-tailed and peaked shape, with significant deviations from normality. The Q-Q plot continues to exhibit strong deviations from the reference line, indicating that the sample size is still insufficient for convergence to the normal distribution.

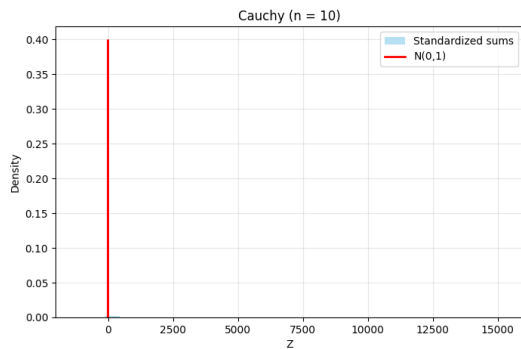


Figure 58:Cauchy (n=10)

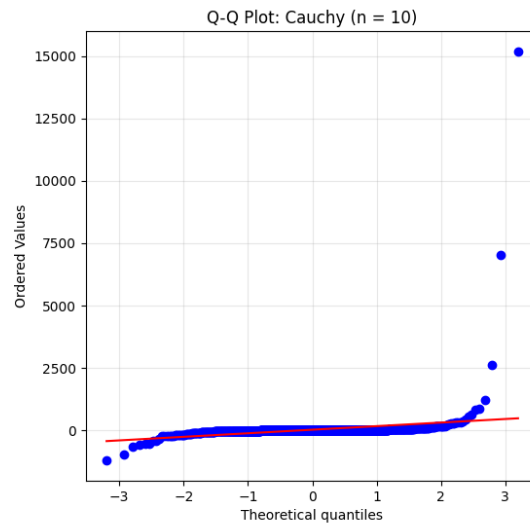


Figure 59:Q-Q Plot Cauchy (n=10)

Explanation for $n = 10$

At $n = 10$, the standardized sums still display non-normal behavior. The histogram remains irregular and heavy-tailed, and the extreme values continue to dominate the distribution. The Q-Q plot shows persistent and strong deviations from the reference line across all quantiles. The Central Limit Theorem does not apply at this sample size.

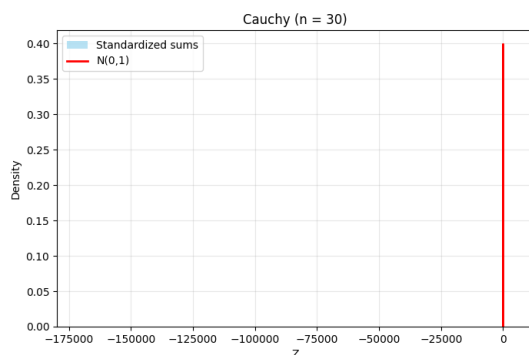


Figure 60: Cauchy (n=30)

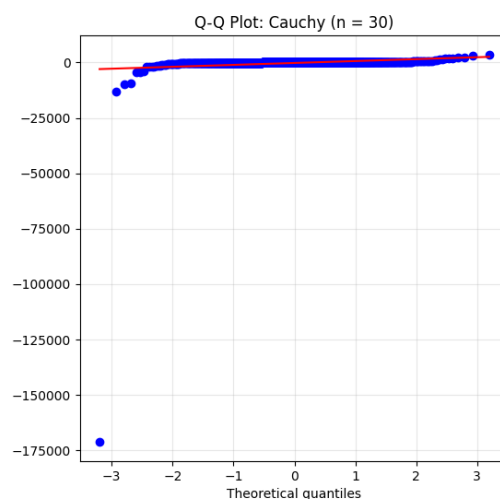


Figure 61:Q-Q Plot Cauchy (n=30)

Explanation for $n = 30$

For $n = 30$, the lack of convergence to the normal distribution is evident. The histogram does not approach a bell-shaped form and remains heavy-tailed and irregular. The Q-Q plot shows extreme deviations from linearity at the tails, confirming that increasing the sample size does not lead to normal convergence for the Cauchy distribution.

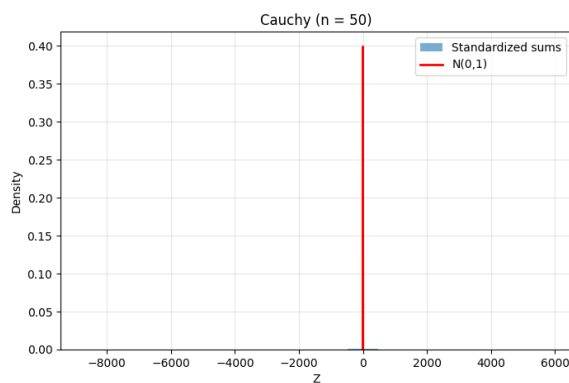


Figure 62: Cauchy ($n=50$)

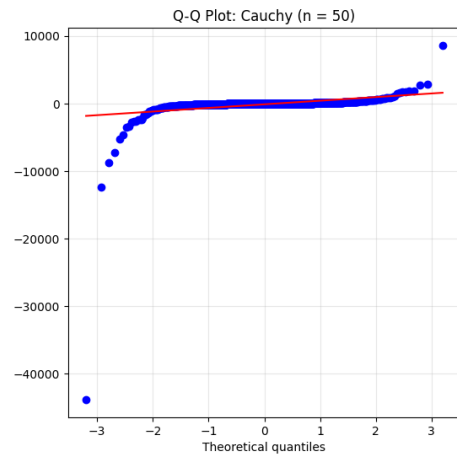


Figure 63: Q-Q Plot Cauchy ($n=50$)

Explanation for $n = 50$

When the sample size reaches $n = 50$, the behavior remains unchanged. The histogram continues to show heavy tails and an irregular shape, and extreme values still affect the distribution. The Q-Q plot shows significant deviations from the reference line. These results highlight the failure of the Central Limit Theorem due to the absence of finite mean and variance.

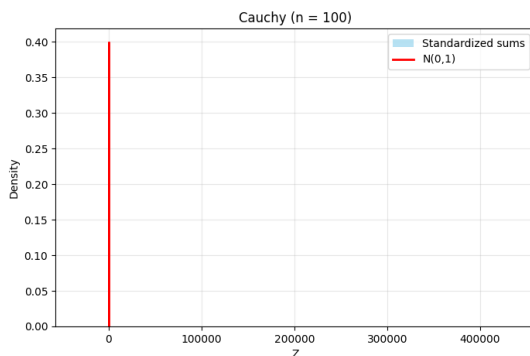


Figure 64: Cauchy ($n=100$)

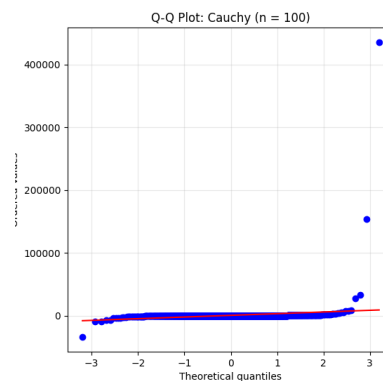


Figure 65 :Q-Q Plot Cauchy ($n=100$)

Explanation for $n = 100$

For $n = 100$, there is still no convergence toward the normal distribution.

The histogram remains heavy-tailed and does not resemble a normal shape. The Q–Q plot exhibits severe deviations from linearity across all quantiles. This confirms that the Central Limit Theorem fails completely for the Cauchy distribution, regardless of the sample size.

6. Conclusion

In this study, the limits of the Strong Law of Large Numbers and the Central Limit Theorem were experimentally investigated using Monte Carlo simulations. By analyzing a variety of probability distributions with different tail behaviors and moment properties, the practical validity of these classical theorems was examined in detail.

The simulation results showed that the Strong Law of Large Numbers holds for distributions with a finite expected value. In such cases, the sample mean eventually stabilizes and converges to the theoretical mean, although the speed and smoothness of convergence depend on the distribution. Light-tailed distributions, such as Uniform and Exponential, exhibit fast and stable convergence, while heavy-tailed distributions, such as Pareto($\alpha = 3$), show slower and noisier convergence.

The Central Limit Theorem was shown to be more restrictive. In addition to a finite mean, CLT requires a finite variance. When this condition is satisfied, convergence to the normal distribution occurs, but the convergence rate is strongly influenced by skewness and tail heaviness. Heavy-tailed distributions require much larger sample sizes to exhibit approximate normality. In cases where the variance is infinite, such as Pareto($\alpha = 1.5$), the Central Limit Theorem fails completely, even though the mean may exist.

The Cauchy distribution represents an extreme example in which neither the Strong Law of Large Numbers nor the Central Limit Theorem applies. The absence of both a finite mean and variance leads to unstable sample means and a complete lack of convergence to the normal distribution. This case clearly demonstrates that probabilistic theorems are not universal and must be applied with careful consideration of their assumptions.

Overall, TW5 highlights the importance of understanding the theoretical conditions behind probabilistic limit theorems. The results show that assumptions such as finite mean and variance are not merely technical details, but essential requirements for meaningful convergence. Through simulation-based analysis, this study provides a deeper and more realistic understanding of the behavior of random variables and reinforces the value of Monte Carlo methods in probability and engineering applications.